

Quantitative Research

Optimal Transport for Calibration of Local Volatility Models

Date: March 30, 2026

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1 Preliminaries

proof KFP

2 Martingale Optimal Transport

Let X be the stochastic process with support D such that for all $t \in [0, 1]$:

$$dX_t = \sigma(t, X_t)dW_t, \quad (1)$$

with initial density p_0 and terminal density p_1 . The KFP equation writes:

$$\begin{cases} \partial_t p - \partial_{xx}(p \cdot \gamma) = 0 & \forall t \in (0, 1) \forall x \in D, \\ p(0, \cdot) = p_0 & \forall x \in D, \\ p(1, \cdot) = p_1 & \forall x \in D, \end{cases} \quad (2)$$

where $\gamma(t, x) = \frac{1}{2}\sigma^2(t, x)$. The associated optimal transport problem consists in finding $\sigma^*(\cdot, \cdot)$ such that:

$$\sigma^* = \arg \min_{\sigma} \int_D \int_0^1 p(t, x) \cdot F(\gamma(t, x)) dx dt, \quad (3)$$

under the constraint that the density of X matches the initial and terminal densities p_0 and p_1 , with F being a convex cost function. According to Strassen's theorem, the problem is feasible if and only if X_0 and X_1 are in convex order, i.e.:

$$\mathbb{E}[\phi(X_0)] \leq \mathbb{E}[\phi(X_1)] \quad (4)$$

$$\Leftrightarrow \int_D \phi(x) p_0(x) dx \leq \int_D \phi(x) p_1(x) dx, \quad (5)$$

for all convex functions ϕ . This condition is easily verified if X is a martingale:

$$\mathbb{E}[\phi(X_0)] = \mathbb{E}[\phi(\mathbb{E}_0[X_1])] \leq \mathbb{E}[\mathbb{E}_0[\phi(X_1)]] = \mathbb{E}[\phi(X_1)], \quad (6)$$

using Jensen inequality.

References

